

Name: _____

TOPIC TEST

Further differentiation (Advanced)

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 10 free-response questions (40 marks)
- Total: 50 marks

Part A

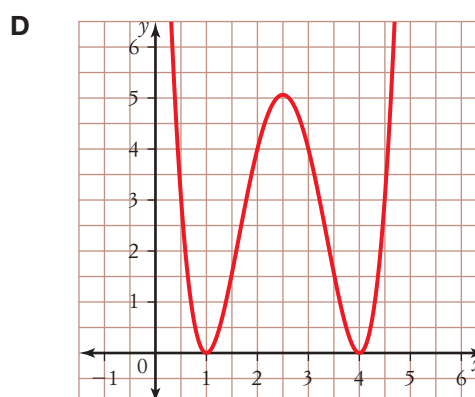
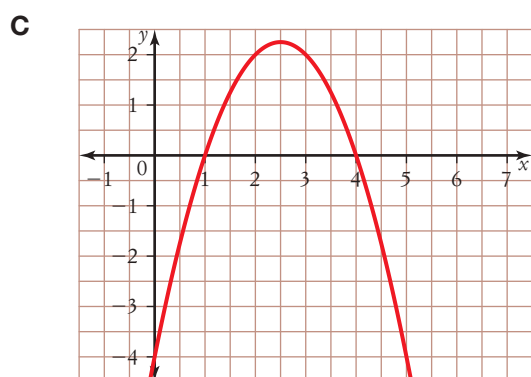
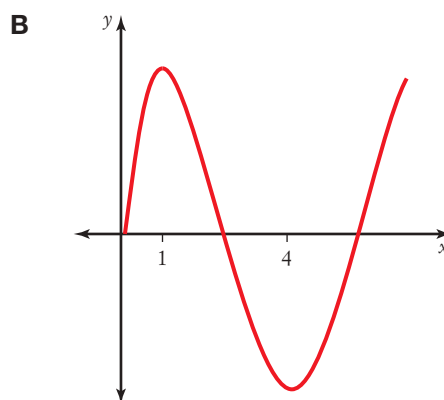
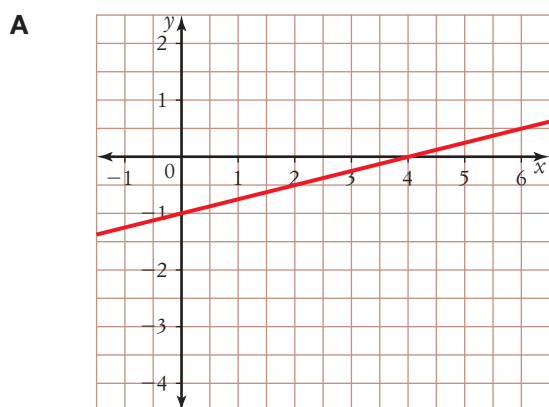
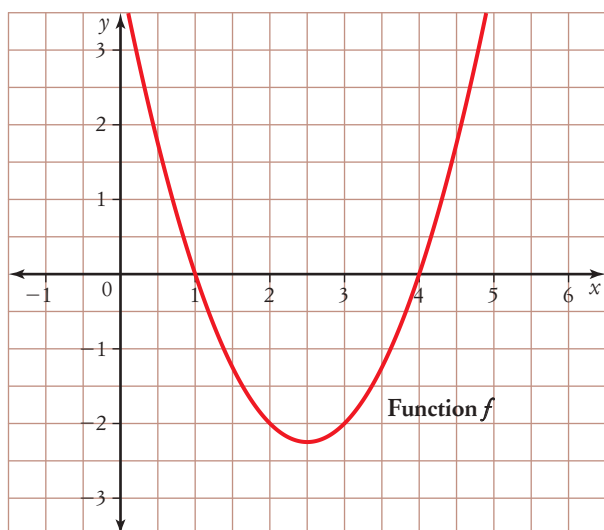
10 multiple-choice questions

1 mark each: 10 marks

Circle the correct answer.

- Find the derivative of $y = x^3 - 3x^2 + 2$.
A $\frac{dy}{dx} = 3x^2 - 6x + 2$ **B** $\frac{dy}{dx} = 3x^2 - 6x$
C $\frac{dy}{dx} = x^2 - 6x$ **D** $\frac{dy}{dx} = 3x^2 - 6$
- Differentiate $y = (x + 1)^2(x - 1)$.
A $\frac{dy}{dx} = (3x - 1)(x + 1)$ **B** $\frac{dy}{dx} = 2x$
C $\frac{dy}{dx} = 2x(x + 1)(x - 1)$ **D** $\frac{dy}{dx} = 2(x + 1)(x - 1)$
- Find the derivative of $f(x) = (5 - 2x^3)^4$.
A $f'(x) = 24x(5 - 2x^3)^3$ **B** $f'(x) = 4x^2(2x^3 - 5)^3$
C $f'(x) = -24x^2(5 - 2x^3)^3$ **D** $f'(x) = -8x(5 - 2x^3)^3$
- Given $y = 3(2x + 7)^4 - 4$, find $\frac{dy}{dx}$.
A $12(2x + 7)^3$ **B** $24(2x + 7)^3$
C $24x^3 - 28$ **D** $12x^3 - 2$
- Find the derivative of $y = e^{\sin x}$.
A $y' = xe^{\sin x}$ **B** $y' = (\sin x)e^{\sin x - 1}$
C $y' = (\cos x)e^{\sin x - 1}$ **D** $y' = (\cos x)e^{\sin x}$
- Find the equation of the tangent to the function $f(x) = 3x^2 - 2x + 1$ at $x = -1$.
A $8x + y + 2 = 0$ **B** $y - 8x + 2 = 0$
C $8y + x - 14 = 0$ **D** $y - 8x - 14 = 0$
- Differentiate $y = \sqrt[3]{\tan x}$.
A $\frac{dy}{dx} = \frac{2}{3\sqrt[3]{\tan x}}$ **B** $\frac{dy}{dx} = \frac{2\sqrt[3]{\tan^2 x}}{3\sec^2}$
C $\frac{dy}{dx} = 2\sec x \sqrt[3]{\tan x}$ **D** $\frac{dy}{dx} = \frac{\sec^2 x}{3\sqrt[3]{\tan^2 x}}$
- Differentiate $y = \frac{\ln(x^2 + 1)}{(x^2 + 1)^4}$.
A $y' = \frac{\ln(x^2 + 1)}{(x^2 + 1)^4}$ **B** $y' = \frac{2x \ln(x^2 + 1)}{(x^2 + 1)^4}$
C $y' = \frac{2x[1 - \ln(x^2 + 1)^4]}{(x^2 + 1)^5}$ **D** $y' = \frac{1 - 4 \ln(x^2 + 1)}{(x^2 + 1)^4}$

- 9 Which graph below is a possible anti-derivative of this graph?



- 10 Find the second derivative of $y = x \sin x^2$.

A $\frac{d^2 y}{dx^2} = 6x \cos x^2 - 4x^3 \sin x^2$

B $\frac{d^2 y}{dx^2} = x \cos x^2 + 4x^3 \sin x^2$

C $\frac{d^2 y}{dx^2} = \cos x^2 - 4x^3 \sin x^2$

D $\frac{d^2 y}{dx^2} = 2x^2 \cos x^2 - \sin x^2$

Part B

10 free-response questions

40 marks

Show your working where appropriate.

11 Find the derivative of each function.

a $f(x) = x^2 - 2x$

b $y = \frac{1}{x}$

c $y = \frac{x}{x^2 - 1}$

d $g(x) = (x^2 - x)^2$

e $h(x) = 4^x$

f $f(x) = 2xe^{x^2}$

g $f(x) = \log_e x^3$

[10 marks]

12 Find the anti-derivative of each function.

a $f(x) = 4x^3 - 3x^2 + x$

b $\frac{dy}{dx} = (5x - 7)^3$

c $g(x) = \tan^2 x \sec^2 x$

d $h(x) = \frac{\sin x}{\cos^2 x}$

e $\frac{du}{dx} = xe^{x^2}$

f $y' = 2x^3 - 2x \sin x^2 \cos x^2$

[10 marks]

13 Find the equation of the normal to the graph for the function $y = x^3 + 2 \ln x^2$ when $x = -1$.

[3 marks]

- 14** A hyperbola has equation $y = \frac{2}{x}$. Find the equation of the tangent to the hyperbola at $x = 2$, in general form.

[4 marks]

- 15** Find the exact gradient of the tangent to the function $f(x) = e^{\ln x^2}$ at $x = \frac{1}{2}$.

[2 marks]

- 16** Find the equation of the function $f(x)$ if $f'(x) = xe^{x^2}$ and the point $P\left(1, \frac{5}{2}\right)$ lies on the graph of the function.

[2 marks]

- 17** Find $y = f(x)$ if $f''(x) = 3x^2 - 6x + 2$, $f'(2) = -3$ and $f(-1) = 2$.

[3 marks]

- 18** Find the derivative of $y = \sin 2x^\circ$.

[1 mark]

19 The rate of change of volume V with respect to time (t) is $\frac{dV}{dt} = (2t-1)^3$.

If $V = 3$ when $t = 0$, find the volume when $t = 3$.

[2 marks]

20 Find the second derivative of $v = \frac{x}{x^2-5}$.

[3 marks]

This is the end of the test.
Use this page for extra working space.

Answers

- | | | | | |
|-----|-----|-----|-----|------|
| 1 B | 2 A | 3 C | 4 B | 5 D |
| 6 A | 7 D | 8 C | 9 B | 10 A |

- 11 a $f'(x) = 2x - 2$ b $y' = -\frac{1}{x^2}$ c $y' = -\frac{x^2+1}{(x^2-1)^2}$
- d $g'(x) = 2(x^2 - x)(2x - 1)$ e $h'(x) = 4^x \ln 4$ f $f'(x) = 2e^{x^2}(2x^2 + 1)$
- g $f'(x) = \frac{3}{x}$
- 12 a $F(x) = x^4 - x^3 + \frac{x^2}{2} + c$ b $y = \frac{(5x-7)^4}{20} + c$ c $G(x) = \frac{\tan^3 x}{3} + c$
- d $y = \ln(\cos^2 x) + c$ e $u = \frac{e^{x^2}}{2} + c$ f $y = \frac{x^4}{2} + \frac{\cos^2(x^2)}{2}$
- 13 $y = x$
- 14 $x + 2y - 4 = 0$
- 15 $m = 1$
- 16 $\frac{e^{x^2}}{2} + 2$
- 17 $f(x) = \frac{x^4}{4} - x^3 + x^2 - 3x - \frac{13}{4}$
- 18 $y' = \frac{\pi}{90} \cos\left(\frac{\pi x^\circ}{90}\right)$
- 19 159
- 20 $\frac{2x^3 + 30x}{(x^2 - 5)^3}$